

HTML Notes — Sinchana

| Complete reference from basics to interview prep

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1. Introduction & Basic Structure

HTML (HyperText Markup Language) is used to structure web pages.

html

```

<!DOCTYPE html>           <!-- Declares HTML5 document type -->
<html lang="en">           <!-- Root element -->
<head>
  <meta charset="UTF-8"/>
  <title>Page Title</title>
</head>
<body>
  <!-- Visible content goes here -->
</body>
</html>

```

Tag	Purpose
<!DOCTYPE html>	Tells browser this is HTML5
<html>	Root of the entire document
<head>	Metadata — not visible on page
<body>	Everything the user sees

2. HTML Tags & Elements

Headings

```

html

<h1>Largest Heading</h1>
<h2>Second Level</h2>
<h3>Third Level</h3>
<!-- ... up to h6 -->

```

Text Tags

```

html

```

```
<p>Paragraph</p>
<span>Inline text</span>
<strong>Bold</strong>
<em>Italic</em>
<br>           <!-- Line break -->
<hr>          <!-- Horizontal line -->
```

Container Tags

```
html

<div>Block container</div>
<span>Inline container</span>
```

Rule: `<div>` is block-level (takes full width). `<span>` is inline (takes only needed width).

3. Attributes

An attribute defines a characteristic of an HTML element. It goes inside the **opening tag** and has two parts: a **name** and a **value**.

```
html

<p align="center">This is centered</p>
<!--   ↑ name     ↑ value           -->
```

Core Attributes (work on most elements)

Attribute	Purpose	Example
<code>id</code>	Unique name for one element	<code><div id="main"></code>
<code>class</code>	Groups multiple elements	<code><p class="highlight"></code>
<code>title</code>	Tooltip on hover	<code><p title="hint"></code>
<code>style</code>	Inline CSS styling	<code><p style="color:red"></code>

Alignment Example

```
html
```

```
<p align="left">Left aligned</p>
<p align="center">Center aligned</p>
<p align="right">Right aligned</p>
```

⚠ Use CSS for alignment in modern HTML. `align` attribute is outdated.

4. Text Formatting

```
html

<strong>Bold text</strong>
<em>Italic text</em>
<u>Underlined</u>
<mark>Highlighted</mark>
<del>Strikethrough</del>
<small>Smaller text</small>
```

Subscript & Superscript

```
html

<!-- Subscript - goes below baseline -->
<p>Water formula: H<sub>2</sub>O</p>

<!-- Superscript - goes above baseline -->
<p>Einstein: E = mc<sup>2</sup></p>
```

Output:

- H₂O
 - E = mc²
-

5. Lists

Unordered List (bullets)

```
html
```

```
<ul>
  <li>Coffee</li>
  <li>Milk</li>
</ul>
```

Output:

- Coffee
- Milk

Ordered List (numbers)

```
html

<ol>
  <li>Coffee</li>
  <li>Milk</li>
</ol>
```

Output:

1. Coffee
2. Milk

Definition List

```
html

<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language – structures web pages.</dd>

  <dt>CSS</dt>
  <dd>Cascading Style Sheets – styles web pages.</dd>

  <dt>JavaScript</dt>
  <dd>Adds interactivity to web pages.</dd>
</dl>
```

Tag	Stands For	Purpose
<code><dl></code>	Definition List	Container for the list
<code><dt></code>	Definition Term	The word/term
<code><dd></code>	Definition Description	The explanation

6. Links & Images

Hyperlink

html

```
<a href="https://www.google.com">Visit Google</a>
<a href="https://www.google.com" target="_blank">Open in new tab</a>
```

Image

html

```

```

Attribute	Purpose
<code>src</code>	Path to the image file
<code>alt</code>	Alternative text (shown if image fails)
<code>width</code>	Width in pixels
<code>height</code>	Height in pixels

7. Audio & Video

Audio

html

```
<audio controls>
  <source src="audio.mp3" type="audio/mpeg">
</audio>
```

Video

```
html

<video controls width="400" height="300" loop autoplay>
  <source src="video.mp4" type="video/mp4">
</video>
```

Attribute

Purpose

controls	Shows play/pause buttons
autoplay	Plays automatically on load
loop	Repeats the video
width / height	Sets dimensions

8. Tables

Basic Structure

```
html

<table border="1">
  <caption>Student Marks</caption>
  <tr>
    <th>Name</th>
    <th>Marks</th>
  </tr>
  <tr>
    <td>Sinchana</td>
    <td>95</td>
  </tr>
</table>
```

Tag	Purpose
<code><table></code>	Creates the table
<code><caption></code>	Title above the table
<code><tr></code>	Table row
<code><th></code>	Table header (bold + centered by default)
<code><td></code>	Table data cell

Caption Alignment (via CSS)

```
html

<caption style="text-align: left;">My Table</caption>
<caption style="text-align: center;">My Table</caption>
```

Styling Two Tables Independently

```
html

<style>
  #table1 { border: 2px solid red; }
  #table1 th { background-color: red; color: white; }

  #table2 { border: 2px solid blue; }
  #table2 th { background-color: blue; color: white; }
</style>

<table id="table1"> ... </table>
<table id="table2"> ... </table>
```

9. Forms

Complete Form Example

```
html
```



```
<form action="/submit.php">

  <!-- Text input -->
  <label for="name">Name:</label>
  <input type="text" id="name" name="name" placeholder="Enter name">

  <!-- Password -->
  <input type="password" placeholder="Enter password">

  <!-- Email -->
  <input type="email" placeholder="Enter email">

  <!-- Radio buttons -->
  <label for="class10">Class 10</label>
  <input type="radio" id="class10" name="class" value="class10">

  <!-- Checkbox -->
  <label for="maths">Maths</label>
  <input type="checkbox" id="maths" name="subject" value="maths">

  <!-- Dropdown -->
  <select name="city" id="city">
    <option value="bangalore">Bangalore</option>
    <option value="mumbai">Mumbai</option>
  </select>

  <!-- Textarea -->
  <textarea name="feedback" placeholder="Give feedback" rows="4" cols="30"></textarea>

  <!-- Buttons -->
  <button type="submit">Login</button>
  <button type="reset">Clear</button>

  <!-- Or using input -->
  <input type="submit" value="Submit">

</form>
```

Form Tags Summary

Tag/Type	Purpose
<code><form></code>	Container for all form elements
<code><label></code>	Describes what the input is for
<code>type="text"</code>	Single line text box
<code>type="password"</code>	Hidden text input
<code>type="email"</code>	Email validation built-in
<code>type="radio"</code>	Choose one from a group
<code>type="checkbox"</code>	Choose multiple options
<code>type="submit"</code>	Submit the form
<code>type="reset"</code>	Clear all fields
<code><select></code>	Dropdown list
<code><textarea></code>	Multi-line text box

10. Semantic HTML

Semantic tags describe their content — they have meaning.

- ✓ `<header>` instead of `<div class="header">`
- ✓ `<nav>` instead of `<div class="navigation">`
- ✓ `<main>` instead of `<div id="content">`
- ✓ `<section>` instead of `<div class="section">`
- ✓ `<article>` instead of `<div class="post">`
- ✓ `<aside>` instead of `<div class="sidebar">`
- ✓ `<footer>` instead of `<div class="footer">`

Page Layout Structure

html

```
<header>
  <h1>Site Title</h1>
  <nav>
    <a href="/">Home</a>
    <a href="/about">About</a>
  </nav>
</header>

<main>
  <section>
    <article>
      <h2>Blog Post Title</h2>
      <p>Published: March 2026 · Author: Sinchana</p>
      <p>Content goes here...</p>
    </article>
  </section>
  <aside>Related links</aside>
</main>

<footer>
  <p>&copy; 2026 Sinchana</p>
</footer>
```

Semantic Tags Reference

Tag	Purpose
<header>	Top section — logo, nav, heading
<nav>	Navigation links
<main>	Primary content (only ONE per page)
<section>	Grouped related content
<article>	Independent, self-contained content
<aside>	Sidebar / related info
<footer>	Bottom section — copyright, links

Benefits of Semantic HTML

-  Better SEO (search engines understand your page)
 -  Better accessibility (screen readers work better)
 -  Cleaner, more readable code
 -  Professional standard in industry
-

11. Multimedia Embedding

Three ways to embed content (PDF, video, maps):

1. `<embed>` — direct inline display

```
html

<embed src="file.pdf" width="500" height="400" type="application/pdf">
```

2. `<iframe>` — window to another page

```
html

<iframe src="https://example.com" width="500" height="400" title="Example"></iframe>
```

Used for: Google Maps, YouTube videos, external websites

YouTube embed: Go to YouTube → Share → Embed → Copy → Paste in your HTML

Google Maps embed: Google Maps → Share → Embed a map → Copy HTML → Paste

3. `<object>` — flexible container

```
html

<object data="file.pdf" width="500" height="400" type="application/pdf">
  Sorry, your browser does not support PDFs.
</object>
```

Tag	Best For
<code><embed></code>	Simple media display
<code><iframe></code>	External pages, maps, YouTube
<code><object></code>	PDFs, plugins, with fallback text

12. Image Maps

Click different parts of one image to open different links.

```
html



<map name="image-map">
  <area
    target="_blank"
    alt="Google"
    title="Google"
    href="http://www.google.com"
    coords="278,104,404,328"
    shape="rect">
</map>
```

⚠ The `src` filename must exactly match the file you saved. Use `alt` so users know what area links to.

13. HTML Entities

Use entities when you need to display special characters.

Essential Entities

Entity	Symbol	Use When
<code>&lt;</code>	<	Inside text (not a tag)
<code>&gt;</code>	>	Inside text

Entity	Symbol	Use When
<code>&amp;</code>	&	Ampersand in text
<code>&quot;</code>	"	Quote inside attribute
<code>&apos;</code>	'	Apostrophe
<code>&nbsp;</code>	(space)	Non-breaking space
<code>&copy;</code>	©	Copyright
<code>&reg;</code>	®	Registered trademark
<code>&trade;</code>	™	Trademark

Math Symbols

Entity	Symbol
<code>&times;</code>	×
<code>&divide;</code>	÷
<code>&plusmn;</code>	±
<code>&ne;</code>	≠
<code>&le;</code>	≤
<code>&ge;</code>	≥

Arrows

Entity	Symbol
<code>&larr;</code>	←
<code>&rarr;</code>	→
<code>&uarr;</code>	↑
<code>&darr;</code>	↓

Example

html

```
<!-- ❌ Wrong -->
```

```
<p>5 > 3 is true</p>
```

```
<!-- ✅ Correct -->
```

```
<p>5 &gt; 3 is true</p>
```

```
<p>&copy; 2026 Sinchana</p>
```

```
<p>Tom &amp; Jerry</p>
```

14. Definition Lists, Subscript & Superscript

Already covered in Sections 5 and 4 — quick recap:

html

```
<!-- Definition List -->
```

```
<dl>
```

```
  <dt>Term</dt>
```

```
  <dd>Description of the term.</dd>
```

```
</dl>
```

```
<!-- Subscript -->
```

```
<p>H<sub>2</sub>O</p>  <!-- H2O -->
```

```
<!-- Superscript -->
```

```
<p>mc<sup>2</sup></p>  <!-- mc2 -->
```

15. Interview Quick Reference

HTML Comments

html

```
<!-- This is a comment — browsers ignore it -->
```

Block vs Inline Elements

Block Elements	Inline Elements
Take full width	Take only needed width
Start on new line	Stay in same line
<code><div></code> , <code><p></code> , <code><h1></code>	<code></code> , <code><a></code> , <code></code>

New in HTML5

```
<header>, <footer>, <nav>, <main>
<article>, <section>, <aside>
<figure>, <figcaption>
<mark>, <time>, <details>, <summary>
<audio>, <video>, <canvas>
<meta viewport> for responsive design
```

Common Interview Questions

Question	Answer
What is <code><!DOCTYPE html></code> ?	Tells browser to use HTML5
Difference between <code>id</code> and <code>class</code> ?	<code>id</code> is unique (one element), <code>class</code> is reusable (many elements)
What is semantic HTML?	Tags that describe their content — <code><header></code> , <code><nav></code> , etc.
<code><div></code> vs <code></code> ?	<code><div></code> is block, <code></code> is inline
What is <code>alt</code> in <code></code> ?	Text shown when image fails to load; helps accessibility
Difference between <code></code> and <code></code> ?	<code></code> has meaning (important), <code></code> is just visual bold
What is a self-closing tag?	Tags with no content: <code>
</code> , <code></code> , <code><input></code> , <code><hr></code>